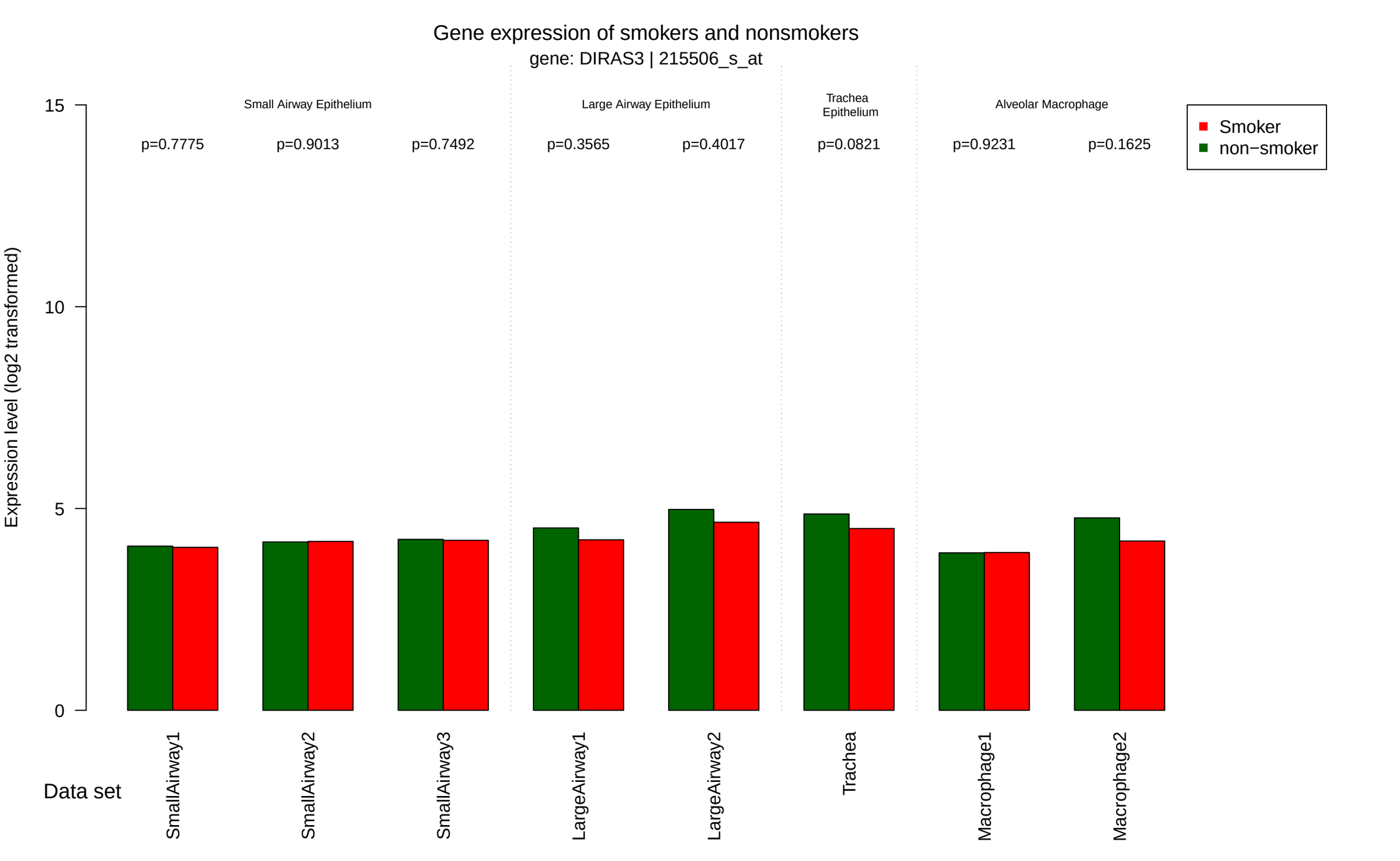
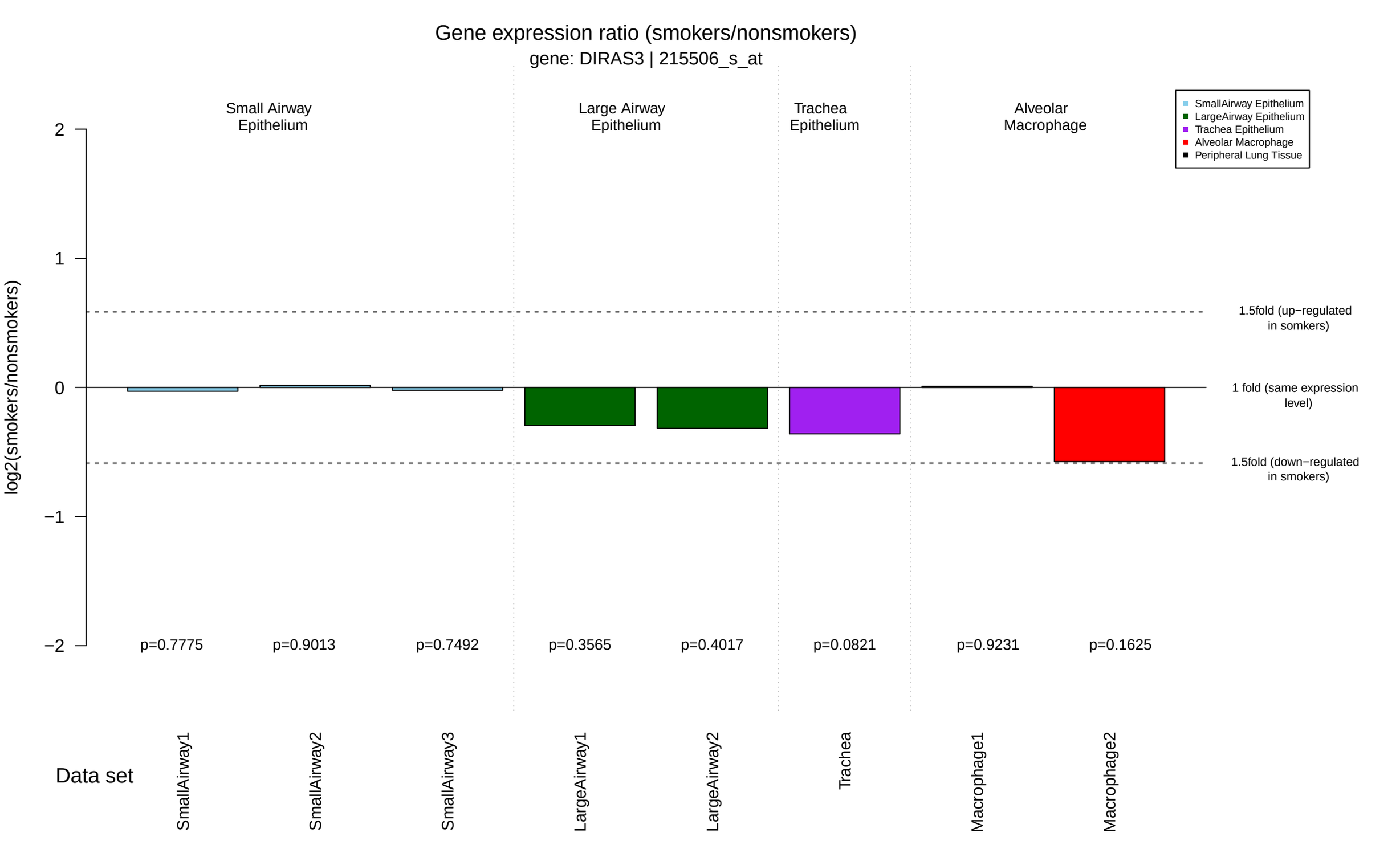
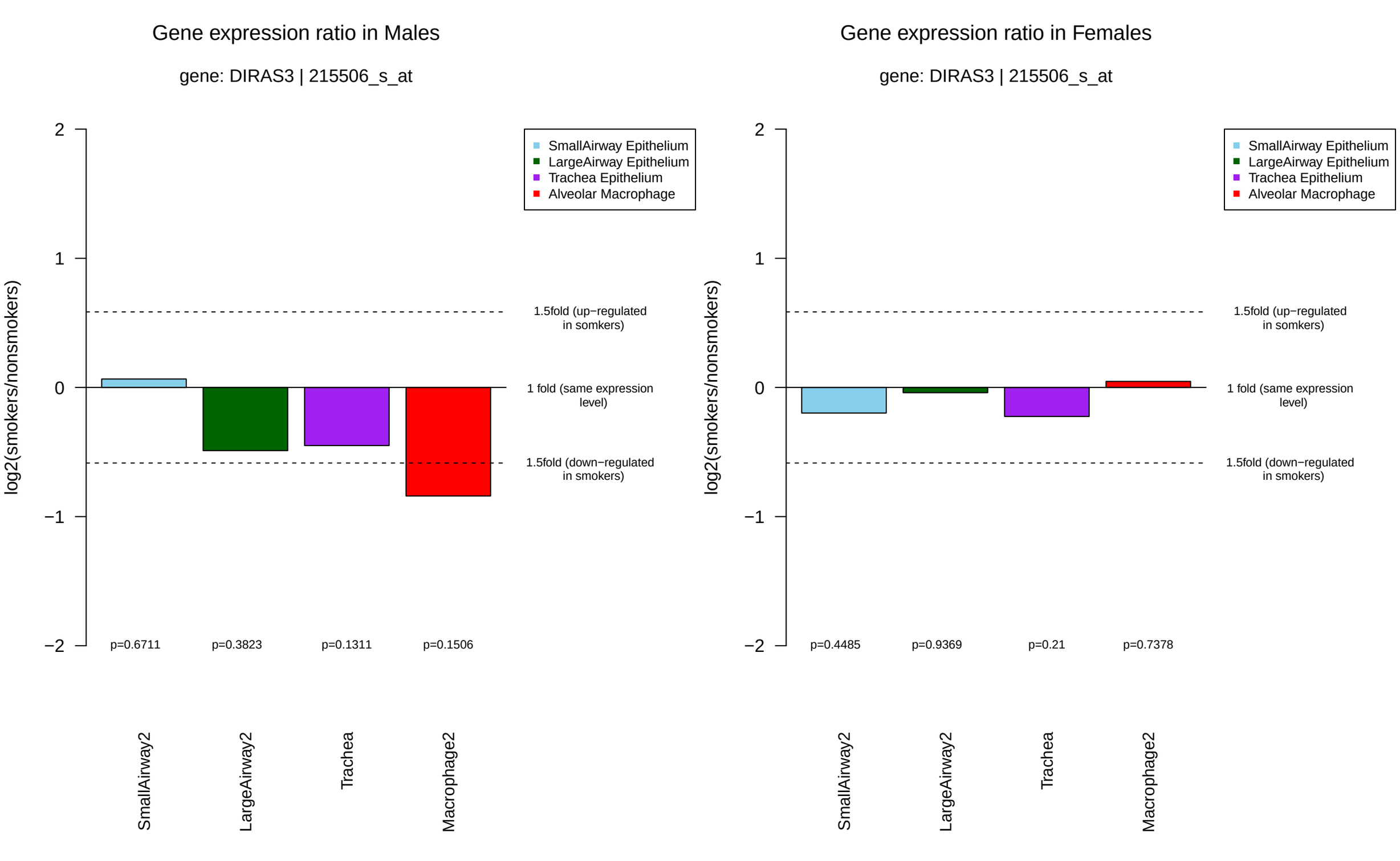
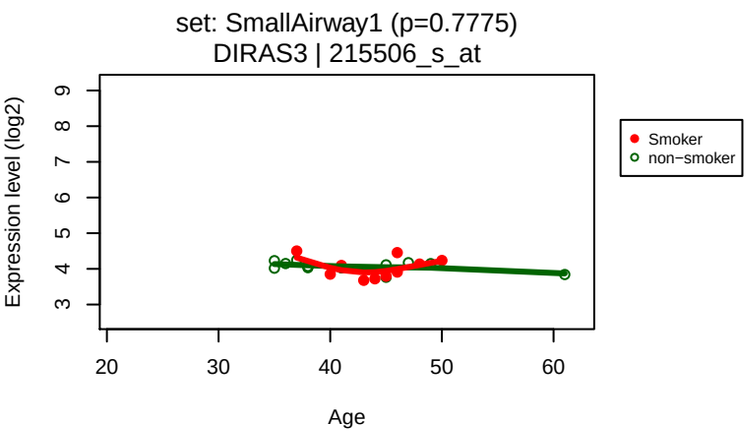
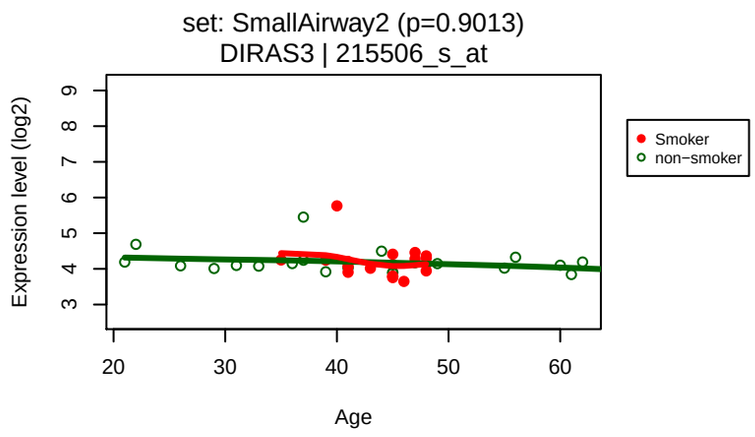
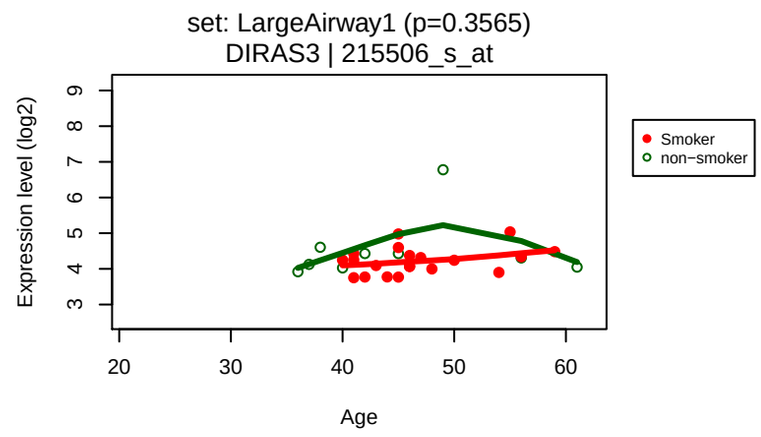

Gene expression of smokers and nonsmokers
gene: DIRAS3 | 215506_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7775
p=0.9013
p=0.7492
p=0.3565
p=0.4017
p=0.0821
p=0.9231
p=0.1625
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

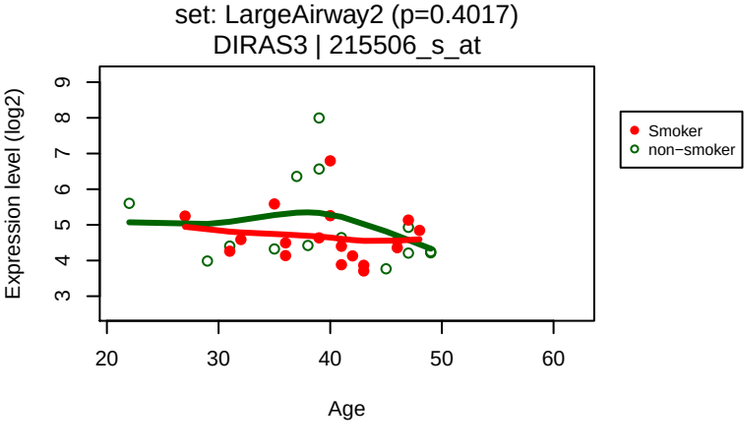

Gene expression ratio (smokers/nonsmokers)
gene: DIRAS3 | 215506_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7775
p=0.9013
p=0.7492
p=0.3565
p=0.4017
p=0.0821
p=0.9231
p=0.1625
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

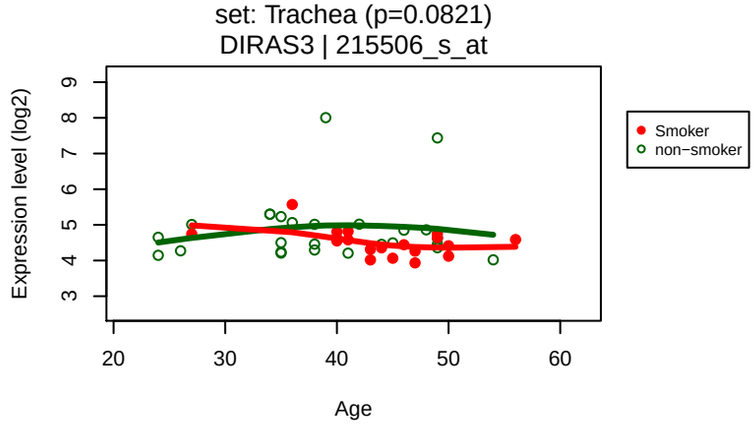

Gene expression ratio in Males
gene: DIRAS3 | 215506_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6711
p=0.3823
p=0.1311
p=0.1506
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DIRAS3 | 215506_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4485
p=0.9369
p=0.21
p=0.7378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

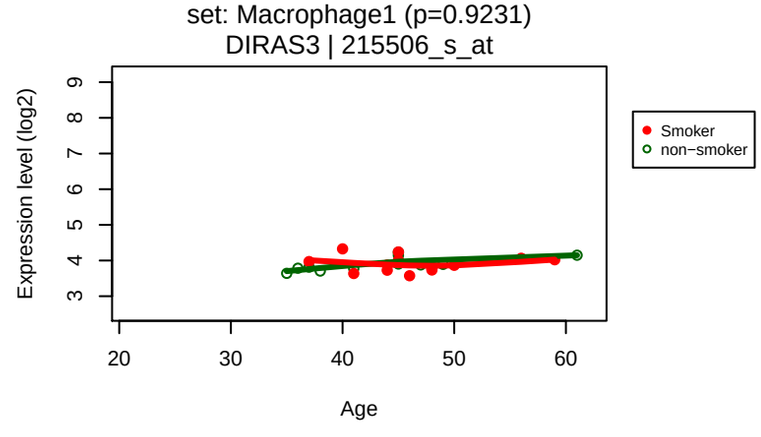

set: SmallAirway1 (p=0.7775)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker

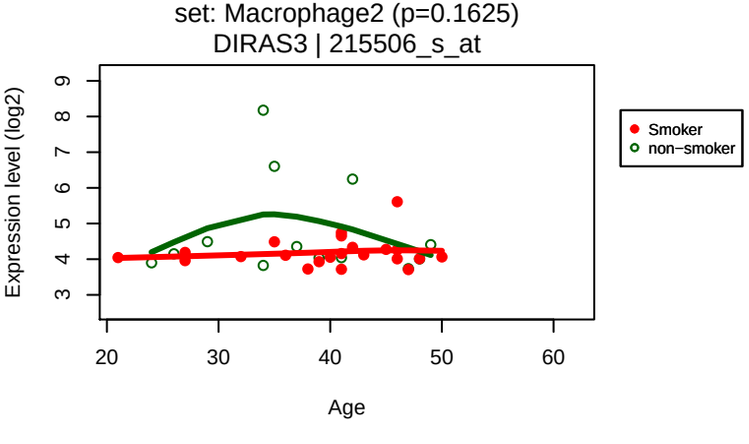

set: SmallAirway2 (p=0.9013)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker

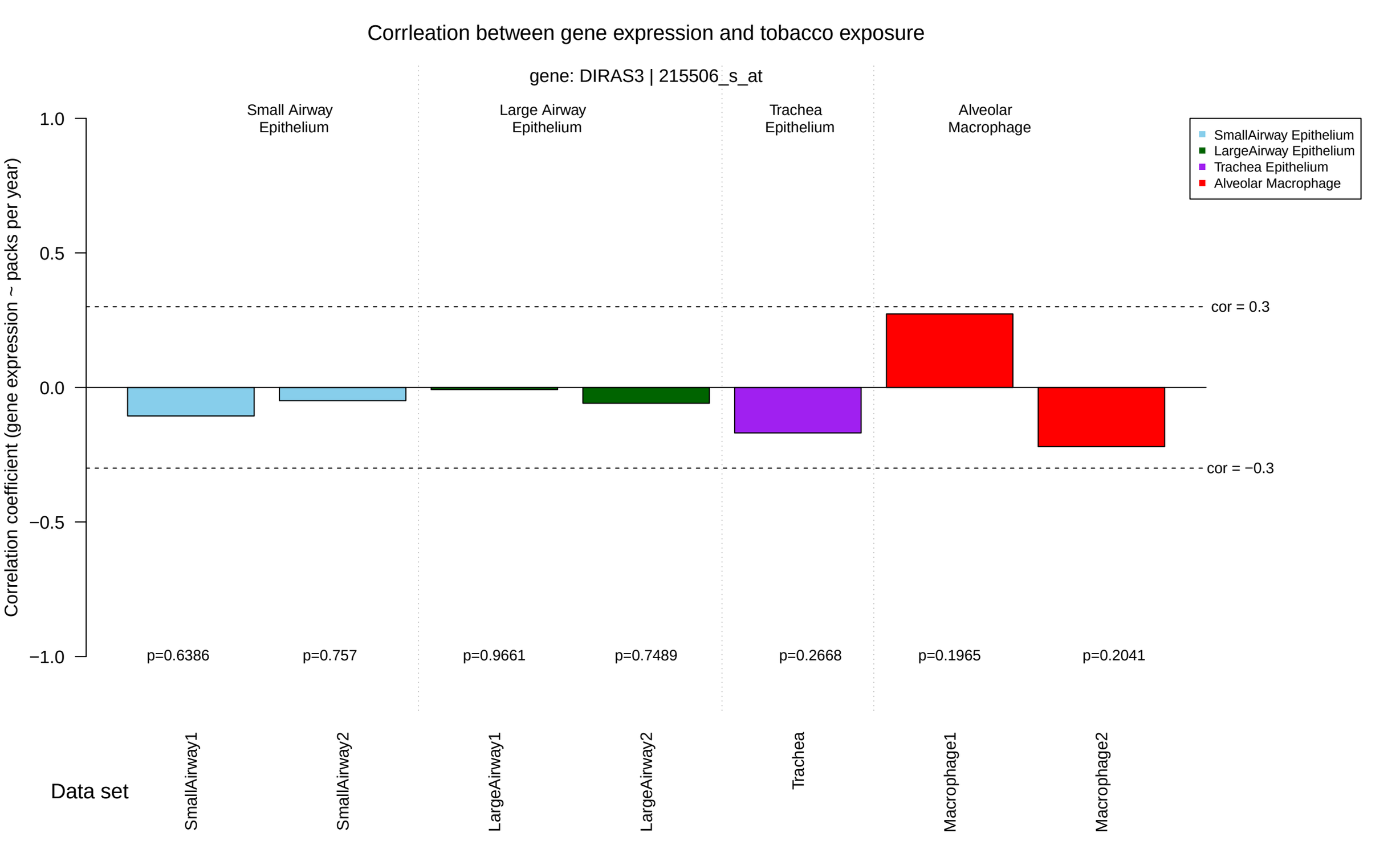

set: LargeAirway1 (p=0.3565)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4017)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0821)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9231)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1625)
DIRAS3 | 215506_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DIRAS3 | 215506_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6386
p=0.757
p=0.9661
p=0.7489
p=0.2668
p=0.1965
p=0.2041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2